

The Skin as an Organ

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The Skin as an Organ – Primary & Secondary Lesions

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Outline

Introduction · Functions of the skin · Anatomy and structure · Morphology of skin lesions · Primary lesions · Secondary lesions · References

Introduction

The skin and its accessory organs – **hair, nails and glands** – are known as the **integumentary system** of the body.

Integument means *covering*, and the skin is the outer covering of the body. It is however **more than a simple body covering**.

The skin is a **complex system of specialized tissues** containing:

- **Glands** that secrete several types of fluids
- **Nerves** that carry impulses
- **Blood vessels** that aid in the regulation of body temperature

The skin is often referred to as the **largest body organ**, and serves as the main **protective barrier** against damage to internal tissues from **trauma, ultraviolet light, temperature, toxins and bacteria**.

Functions of the skin

(i) **Protection.** As a protective membrane over the entire body, the skin guards the deeper tissues against **excessive loss of water, salts and heat**, and against **invasion of pathogens and their toxins**.

(ii) **Secretion.** The skin contains **two types of glands**:

- **Sebaceous glands** produce an oily secretion called **sebum**, which helps lubricate the surface of the skin
- **Sweat glands** produce a watery secretion called **sweat**, which helps cool the body as it evaporates from the skin surface

(iii) **Sensation.** **Nerve fibres** located under the skin act as receptors for sensations such as **pain, temperature, pressure and touch**.

(iv) **Thermoregulation.** Several different tissues in the skin aid in **maintaining the body temperature**.

Anatomy / structure of the skin

The three layers of the skin, and the five layers of the epidermis

There are **three layers** in the skin. From the outer surface inwards:

1. **Epidermis** – a thin cellular membrane layer
2. **Corium or dermis** – dense, fibrous connective tissue layer
3. **Subcutaneous tissue or hypodermis** – thick, fat-containing tissue

Skin structure showing the glands, hair follicle and vascular plexus

The epidermis

Cells move **from the base of the epidermis up to the surface**, changing shape and structure as they go.

The epidermis is divided into **five layers**, from upper to lower:

i. **Stratum corneum** – made up of **stratified squamous epithelium** or hardened cells, which play a role in the skin's **protective function**.

ii. **Granular layer** – the **transitional layer** where keratinocyte skin cells develop into their final form and die. It contains **keratohyalin** and **lamellar granules**.

iii. **Spinous layer** – **keratin-producing epidermal cells** that provide a continuous **net-like layer of protection** for underlying tissue.

iv. **Basal layer** – contains special cells called **melanocytes**. Melanocytes form and contain a black pigment called **melanin**.

The number of melanocytes in all races is the same – but the **amount of melanin** accounts for the colour differences among the races.

v. **Basement membrane** – **connects and functionally separates** the epidermis and the dermis.

The epidermis contains **no blood vessels, no lymphatic vessels and no connective tissue** (elastic fibres, cartilage, fat). It is therefore **dependent on the deeper dermis** and its rich network of capillaries for nourishment.

The dermis

Composed of **blood and lymph vessels and nerve fibres**, as well as the **accessory organs of the skin** – the **hair follicles, sweat glands and sebaceous glands**.

Main function: to provide **physical support and nutrients to the epidermis**.

Two layers are identified within the dermis:

- **Papillary dermis** – contains **smaller blood vessels** which supply oxygen, elastic fibres and nutrients to the **lower epidermis**
- **Reticular dermis** – the **thicker** layer; contains **dense connective tissue, larger blood vessels, elastic fibres, and bundles of collagen arranged in layers**

Key cell types within the reticular layer:

- **Fibroblasts** – a key cell involved in **repairing tissue damage**
- **Mast cells** – involved in **fighting infection**
- **Lymphatic vessels** – a key part of the body's defence against infection
- **Epidermal appendages or rete pegs** – link the epidermis and dermis together to **prevent skin damage**
- **Ground substance** – a **gel-like substance** that supports the cells within the dermis and provides structure to the area

The subcutaneous layer / hypodermis

Composed of **adipose (fat) tissue**. Important in **protection of the deeper tissues** of the body, and as a **heat insulator**.

Morphology of skin lesions

A **skin lesion** refers to **any skin area that has different characteristics from the surrounding skin** – including **colour, shape, size and texture**.

Skin lesions are very common and often appear as a result of **localized damage to the skin**, like sunburns or contact dermatitis.

Others, however, can be **manifestations of underlying disorders**, such as diabetes, infections etc.

Lesions are classified into **two**:

- **Primary lesions** – originate on **previously healthy skin** and are **directly associated with a specific cause**. Examples: bullae, burrow, macule, nodule, papule, plaque, pustule, tumour, vesicle and wheal
- **Secondary lesions** – develop from the **evolution of a primary skin lesion**, either due to **traumatic manipulation** (such as scratching or rubbing), or due to its **treatment or progression**. Examples: crusts, sores, ulcers, scars etc.

Primary skin lesions

Vesicle and bulla – raised lesions filled with clear fluid

Lesion	Definition
Vesicle	A raised lesion filled with clear fluid , < 1 cm in diameter
Bulla	A raised lesion filled with clear fluid , > 1 cm in diameter
Papule	A circumscribed palpable elevation with distinct borders, < 0.5 cm in diameter
Plaque	An elevated area of skin with distinct borders and an epidermal change , > 1 cm in diameter
Macule	A circumscribed alteration in the colour of the skin that is flat to the surface and not palpable , < 1 cm in size
Patch	A large macule , > 1 cm in size
Nodule	A circumscribed raised solid lesion , ≤ 1 cm
Wheal	A transient area of dermal (or dermal and hypodermal) oedema – compressible and usually evanescent
Burrow	A small linear tunnel in the skin that houses a parasite such as scabies
Pustule	A circumscribed elevation of pus-filled skin lesion, < 1 cm

Papule and plaque

Macule and patch – flat, not palpable

Macules and patches on skin

Nodules – circumscribed raised solid lesions

Wheals – transient, compressible dermal oedema

Burrow – a small linear tunnel housing a parasite

Secondary skin lesions

Lesion	Definition
Crust	A hard, friable, irregular layer of dried blood, serum, pus, tissue debris or any combination of these, adherent to the surface of injured or inflamed skin
Ulcer	An open sore or erosion of the skin or mucous membrane
Scaling	Whitish plates or flakes of stratum corneum present on the skin surface
Scar	Fibrous tissue replacing normal tissue destroyed by injury or disease

Lesion	Definition
Erosion and oozing	Moist, circumscribed, slightly depressed areas representing a blister base with the roof of the blister removed
Desquamation	The peeling of sheets of scale after an acute injury to skin (burns, toxic drug reaction)
Excoriation	Oval to linear depressions in the skin with complete removal of the epidermis , exposing a broad section of the red dermis (atopic dermatitis)
Fissure	Linear, wedge-shaped cracks in the epidermis extending down to the dermis and narrowing at the base (warts)
Lichenification	A thickening of the epidermis (and to some extent the dermis) in response to prolonged rubbing
Gangrene	Death of tissue , usually due to loss of blood supply

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